1 Two blocks, A and B, are joined by a thin thread that passes over a frictionless pulley. Block A is at rest on a rough horizontal surface and block B is held at rest, just below the pulley.

Fig. 1.1 shows the thread hanging loose.

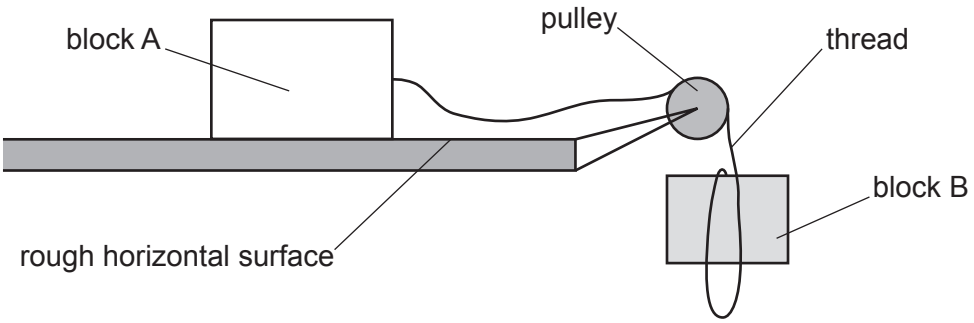


Fig. 1.1 (not to scale)

Block B is released and it falls vertically. The thread remains loose until block B has fallen a distance of 0.45 m.

The mass of block B is 0.50 kg.

(a) Calculate the change in the gravitational potential energy (g.p.e.) of block B as it falls through 0.45 m.

change in g.p.e. ..... [2]

(b) The mass of block A is 2.0 kg.

When the thread tightens, it pulls on block A which moves to the right at a speed of 0.60 m / s.

(i) Calculate the impulse exerted on block A as it accelerates from rest to 0.60 m / s.

impulse = ... [3]

(ii) Both of the blocks now move at a constant speed of 0.60 m / s until block B hits the ground and the thread becomes loose.

Explain the energy change that takes place in block A after block B stops moving.

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..... [3]

[Total: 8]